

Transfer Learning Homework Report

Introduction

In this assignment, I experimented with transfer learning using pretrained DenseNet201 and ResNet50 models on the UCMerced Land-use and Intel Image Classification datasets. I compared three data augmentation schemes—baseline (no augmentation), Augmentation A (geometric + brightness/contrast), and Augmentation B (color shift + noise/blur)—across varying training set sizes ($n = 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1000$). Models were fine-tuned for 5 epochs, freezing the backbone and training only the final classifier layer. MixUp and CutMix were not applied in this experiment.

Results Summary

The following figures show the training, validation, and test accuracy curves for each model and dataset combination. Please refer to the embedded plots for details.

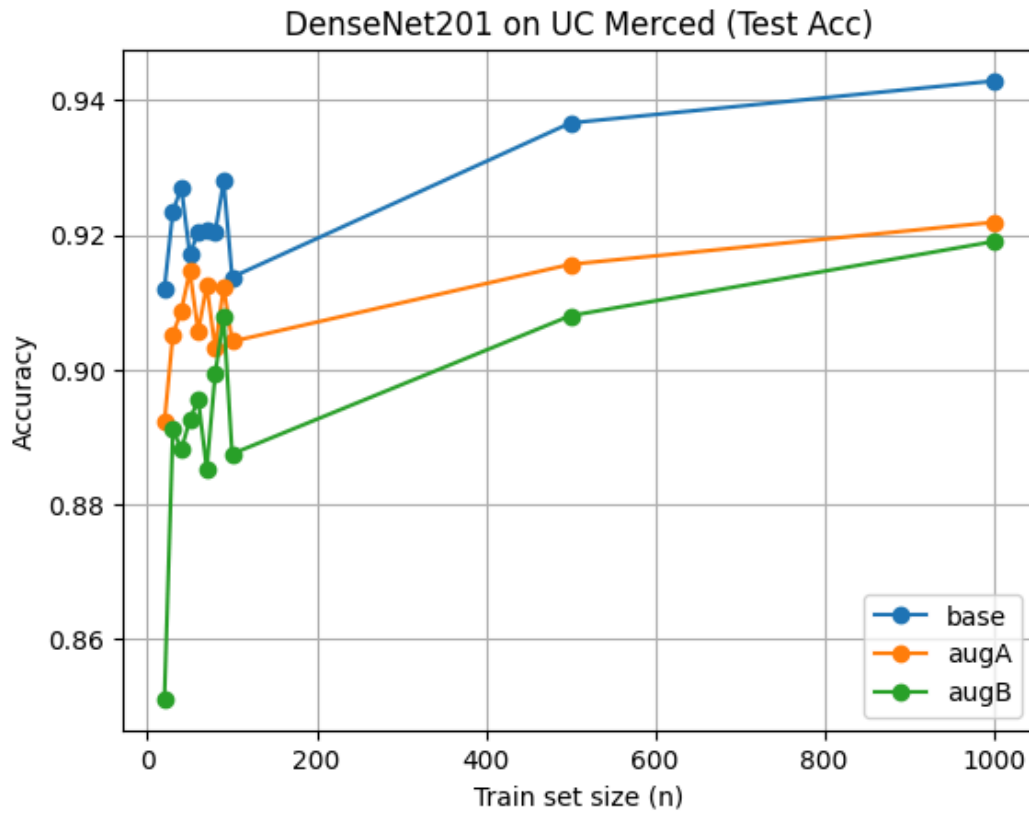


[GitHub](#)

[Neural-Network-HW4-Transfer-Learning](#)

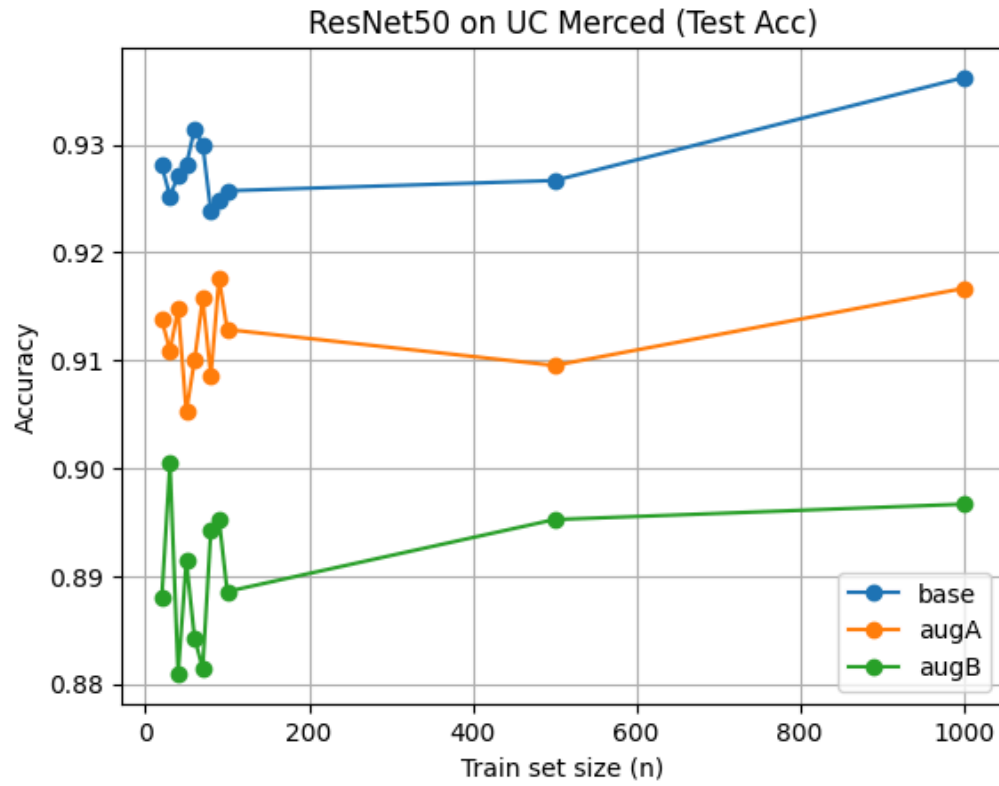
DenseNet201 on UCMerced

Figure: DenseNet201 on UCMerced accuracy curves (train / val / test) under the three augmentation schemes.



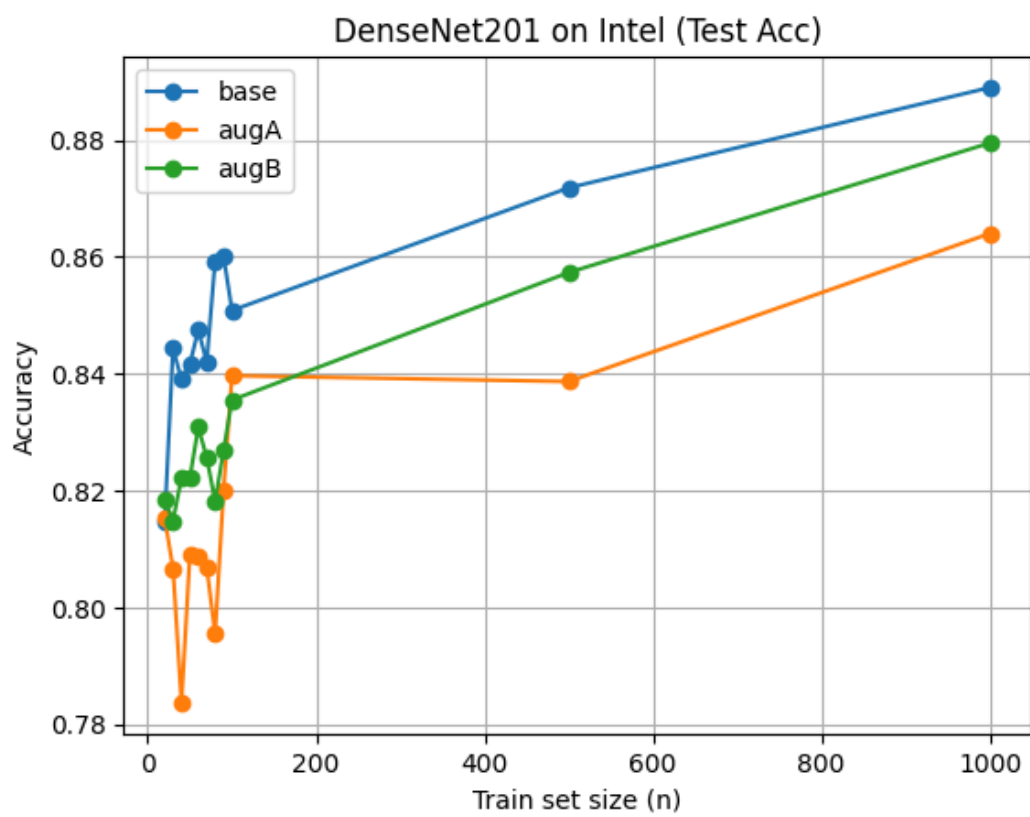
ResNet50 on UCMerced

Figure: ResNet50 on UCMerced accuracy curves (train / val / test) under the three augmentation schemes.



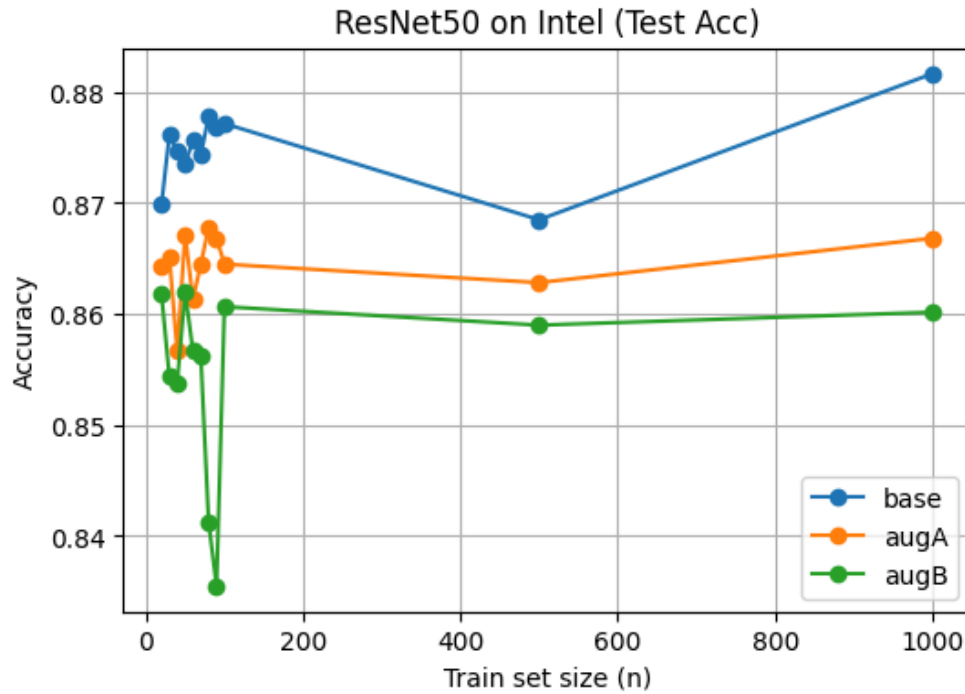
DenseNet201 on Intel

Figure: DenseNet201 on Intel accuracy curves (train / val / test) under the three augmentation schemes.



ResNet50 on Intel

Figure: ResNet50 on Intel accuracy curves (train / val / test) under the three augmentation schemes.



Discussion

Overall, the baseline (no augmentation) achieved the highest test accuracy, particularly at larger training sizes. Augmentation A provided moderate benefits in low-data regimes and narrowed the gap as the amount of training data increased. Augmentation B started behind but showed significant gains with larger datasets. DenseNet201 slightly outperformed ResNet50 in baseline settings, though both architectures followed similar trends under varying data volumes and augmentations.

Conclusion

These experiments demonstrate the importance of data volume and augmentation strategy in transfer learning. Future extensions could include MixUp and CutMix augmentations, as well as longer fine-tuning schedules or unfreezing additional layers.

